Memory management bits in `arch/`

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When arch/ meets mm/

- TLB management
- Page table manipulations
- Memory models
- Memory detection and initialization
  - Cold and hot (un)plug

Short quiz

`pgd_offset(mm, address)`

returns a pointer to the entry in the top-level page table that maps the address.

Linux supports ~~25~~ 24 architectures.

a) How many definitions of `pgd_offset` does Linux have?
b) How many does it really need?

Short quiz

A. 31
B. 24
C. 2
D. 1



Page table manipulation

- Folding is neat, but…
  - Lots of “empty” lines
  - Tree-wide updates for each new level (once in couple of years)
- Possible alternatives:
  - Use page walk
  - Completely new interface, e.g

```
vpte_for_each(vpte, start, end, flags)
```

- Split and clean `asm/page.h` and `asm/pgtable.h`
  - For instance, like x86...



Memory models

- DISCONTIGMEM is still with us
  - alpha, arc, ia64, m68k

commit d41dee369bff3b9dcb6328d4d822926c
Date: Thu Jun 23 00:07:54 2005 -0700

[PATCH] sparsemem memory model

Sparsemem replaces DISCONTIGMEM when enabled, and it is hoped that it can eventually become a complete replacement

SPARSEMEM

- alpha
  - Nobody cares?
- IA-64
  - Nobody cares?
  - Weird dependency of SPARSEMEM on DISCONGITMEM
  - Custom VMEMMAP implementation (VIRTUAL_MEM_MAP)



SPARSEMEM

- arc
  - Highmem is below lowmem
  - Less efficient than DISCONTIGMEM?
  - Wasted mem_map
- m68k
  - Unknown memory bank configurations
  - Wasted mem_map



SPARSEMEM

- Decrease section size
  - Increased `mem_section[]` size
  - Descreased `MAX_PHYSMEM_BITS`
- Use `VMEMMAP`
  - More memory overhead
- Cut unused memmap when `HAVE_ARCH_PFN_VALID=y`
  - Requires custom pfn_valid()
    - Already exists on m68k
    - May use memblock, like arm/arm64

Memory initialization

- Various amount of cruft to get from firmware memory info to functional buddy allocator
- Several `[start, end]` constructs to represent memory banks
  - some architectures have more than one
- No consistency in treatment of the reserved areas
  - Is it memory or not?

Vision

- Early reservations and memory detection happen **before** `setup_arch()`
- Memory detection determines NUMA configuration
- Unified representation of (coldplug) memory layout



Memory layout in memblock

- Keep allocator `__init` only, retain data and accessors
  - Half way there for s390 physmem
- x86: is reserved memory also memory?
- Use ARM's pfn_valid() for systems with SPARSEMEM and few memory banks
- Extend to memory hotplug?



Challenges

- Hard to get feedback from less active architectures
- Non-trivial changes are scary
- No consideration for neighbours for new arch/ code



Thank you!

RFC: reserve and detect memory early

```
diff --git a/init/main.c b/init/main.c
@@ -595,4 +595,6 @@ asmlinkage void start_kernel(void)
        pr_notice("%s", linux_banner);
+       memory_reserve_early();
+       memory_detect();
        setup_arch(&command_line);
        mm_init_cpumask(&init_mm);
        setup_command_line(command_line);
```

RFC: dissolve `mem_init()`

```
diff --git a/init/main.c b/init/main.c
@@ -556,6 +558,7 @@ static void __init mm_init(void)
        page_ext_init_flatmem();
+       memblock_free_all();
+       free_highmem_pages();
        report_meminit();
-       mem_init();
        kmem_cache_init();
        pgtable_init();
@@ -566,4 +569,5 @@ static void __init mm_init(void)
        init_espfix_bsp();
-       pti_init();
+       arch_post_mm_init();
}
```